

Toshinori Kitamura

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Positions

University of Alberta

POSTDOCTORAL RESEARCHER

Sep 2025 – Present

- Supervisor: Prof. Csaba Szepesvári

The University of Tokyo

POSTDOCTORAL RESEARCHER

Apr 2025 – Sep 2025

- Supervisor: Prof. Yutaka Matsuo

Education

The University of Tokyo

PH.D. IN ENGINEERING

2022 – 2025

- Supervisors: Dr. Tadashi Kozuno and Prof. Yutaka Matsuo

Nara Institute of Science and Technology (NAIST)

M.S. IN SCIENCE AND TECHNOLOGY

2020 – 2022

- Supervisor: Prof. Takamitsu Matsubara

Keio University

B.E. IN SCIENCE AND TECHNOLOGY

2016 – 2020

- Supervisor: Prof. Toru Namerikawa

University of California, Davis

EXCHANGE STUDENT

2018 – 2019

- Supervisor: Prof. Lifeng Lai

Publications

Conference and Journal Papers

- “Revisiting Subgradient Dominance in Robust MDPs: Counterexamples, Hardness, and Sufficient Conditions”
Toshinori Kitamura, Arnob Ghosh, Alex Ayoub, Thang D. Chu, Csaba Szepesvári, **NeurIPS 2026 (acceptance rate 25.7%)**
- “Randomized Exploration for Linear Bandits via Absolute Perturbations”
Toshinori Kitamura, Shuai Liu, Csaba Szepesvári, **RLC 2026 (acceptance rate 33.9%)**
- “Offline-to-Online Learning in Linear Bandits”
Kushagra Chandak, **Toshinori Kitamura**, Xiaoqi Tan, **RLC 2026 (acceptance rate 33.9%)**
- “Emergence of Exploration in Policy Gradient Reinforcement Learning via Retrying”
Soichiro Nishimori, Paavo Parmas, Sotetsu Koyamada, Tadashi Kozuno, **Toshinori Kitamura**, Shin Ishii, Yutaka Matsuo, **ICML 2026 (acceptance rate 26.6%)**
- “Provably Efficient RL under Episode-Wise Safety in Constrained MDPs with Linear Function Approximation”
Toshinori Kitamura, Arnob Ghosh, Tadashi Kozuno, Wataru Kumagai, Kazumi Kasaura, Kenta Hoshino, Yohei Hosoe, Yutaka Matsuo, **NeurIPS 2025 (Spotlight, top 3.2%; acceptance rate 24.5%)**
- “Near-Optimal Policy Identification in Robust Constrained Markov Decision Processes via Epigraph Form”
Toshinori Kitamura, Tadashi Kozuno, Wataru Kumagai, Kenta Hoshino, Yohei Hosoe, Kazumi Kasaura, Masashi Hamaya, Paavo Parmas, Yutaka Matsuo, **ICLR 2025 (acceptance rate 31.9%)**
- “Regularization and Variance-Weighted Regression Achieves Minimax Optimality in Linear MDPs: Theory and Practice”
Toshinori Kitamura, Tadashi Kozuno, Yunhao Tang, Nino Vieillard, Michal Valko, Wenhao Yang, Jincheng Mei, Pierre Ménard, Mohammad Gheshlaghi Azar, Rémi Munos, Olivier Pietquin, Matthieu Geist, Csaba Szepesvári, Wataru Kumagai, Yutaka Matsuo, **ICML 2023 (acceptance rate 27.9%)**
- “Geometric Value Iteration: Dynamic Error-Aware KL Regularization for Reinforcement Learning”
Toshinori Kitamura, Lingwei Zhu, Takamitsu Matsubara, **ACML 2021 (acceptance rate 30.4%)**
- “Cautious Actor-Critic”
Lingwei Zhu, **Toshinori Kitamura**, Takamitsu Matsubara, **ACML 2021 (acceptance rate 30.4%)**

Workshop Papers, Preprints, and Domestic Conferences

- “A Unified MDP Framework for Solving Robust, Convex, Multi-Discount Constraints, and Beyond”
Toshinori Kitamura, Arnob Ghosh, Tadashi Kozuno, Kenta Hoshino, Yohei Hosoe, Kazumi Kasaura, Wataru Kumagai, Paavo Parmas, Yutaka Matsuo, **Finding the Frame Workshop at RLC 2025**
- “A Policy Gradient Primal-Dual Algorithm for Constrained MDPs with Uniform PAC Guarantees”
Toshinori Kitamura, Tadashi Kozuno, Masahiro Kato, Yuki Ichihara, Soichiro Nishimori, Akiyoshi Sannai, Sho Sonoda, Wataru Kumagai, Yutaka Matsuo, **RLC Safety Workshop 2024 (Oral)**
- “On Various Problem Settings of Sequential Decision Making and How Prior Knowledge of Problems Affects Theoretical Guarantees”
Tadashi Kozuno, **Toshinori Kitamura**, Yuki Ichihara, Makoto Hagiwara, **JSAI 2023**
- “KL-Entropy-Regularized RL with a Generative Model is Minimax Optimal”
Tadashi Kozuno, Wenhao Yang, Nino Vieillard, **Toshinori Kitamura**, Yunhao Tang, Jincheng Mei, Pierre Ménard, Mohammad Gheshlaghi Azar, Michal Valko, Rémi Munos, Olivier Pietquin, Matthieu Geist, Csaba Szepesvári, **arXiv 2022**
- “ShinRL: A Library for Evaluating RL Algorithms from Theoretical and Practical Perspectives”
Toshinori Kitamura, Ryo Yonetani, **NeurIPS Deep RL Workshop 2021**

Teaching

Foundations of RL, The University of Tokyo, Japan

2027

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- Slides

Deep RL Spring Seminar, The University of Tokyo, Japan

2023, 2024, 2025

LECTURER

- Course page

Deep Learning Course, The University of Tokyo, Japan

2023, 2024, 2025

LECTURER

- Course page

Awards

- **Open Champion, 2026:** Alberta Provincial Kendo Championships
- **NeurIPS 2025 Spotlight** (688 of 21,575 submissions, top 3.2%): Provably Efficient RL under Episode-Wise Safety in Constrained MDPs with Linear Function Approximation
- **Head of Department's Award (Research), 2025:** Department of Technology Management for Innovation, Graduate School of Engineering, The University of Tokyo

Professional Service

- **Area Chair:** ICLR
- **Reviewer:** NeurIPS, ICLR, RLC

Research Experience

RIKEN Center for Advanced Intelligence Project

2023 – 2025

PART-TIME RESEARCHER, DEEP LEARNING THEORY TEAM

- Statistical RL theory: constrained RL and general function approximation

OMRON SINIC X Corporation

2021 – 2022, 2023 – 2024

RESEARCH INTERN

- RL library with JAX and statistical RL theory

National Institute of Advanced Industrial Science and Technology (AIST)

2019 – 2020

RESEARCH INTERN

- RL for mobile robot navigation in human crowds

Industry Experience

Integral AI, Inc.

2022

ENGINEERING INTERN (ROBOTICS)

ugo, Inc.

2018

ENGINEERING INTERN (ROBOTICS)